UE26CY151A · ENGINEERING CHEMISTRY

SEMESTER 1 · UNIT 1

00

MASTER EDITION

Unit 1 — Complete Study Pack

Everything in one place: notes, formulas, figures, flowcharts, worked examples and past-paper questions, organised topic by topic.

M1
Molecular Spectroscopy
8 topics

M2
Phase Equilibria
5 topics

M3
Computational Chemistry
1 topic

Exam weight	★★★ high · ★★☆ medium · ★☆☆ low
Flags	NOT ON CURRENT SLIDES · TRANSCRIPT ONLY · ADDED FOR CLARITY
Sources	S004·8 slide 8 of deck 004 · T009 class transcript · PDF Unit-1 slide PDF · ESA past paper

Compiled from PESU AV-summary transcripts, class slides and past ESA papers · 13 October 2026

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T2 Electromagnetic spectrum & regions ★★★

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T3b Beer–Lambert law ★☆☆

T4 Rotational (microwave) spectroscopy ★★★

T5 Vibrational (IR) – SHO ★★★

T6 Vibrational – anharmonic ★★☆

T7 Electronic spectroscopy & Franck–Condon ★★★

M2 Phase Equilibria

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MODULE 1

Molecular Spectroscopy

T1 · EM radiation & interaction with matter | T2 · Electromagnetic spectrum & regions | T3 · Born–Oppenheimer approximation | T3b · Beer–Lambert law | T4 · Rotational (microwave) spectroscopy | T5 · Vibrational (IR) – SHO | T6 · Vibrational – anharmonic | T7 · Electronic spectroscopy & Franck–Condon

T1 EM radiation & interaction with matter ★★★

SHORT NOTE

Spectroscopy studies radiation–matter interaction. A spectrophotometer has source, monochromator, sample, detector, recorder. EM waves have E ⊥ B ⊥ direction. E = hν, ν = c/λ, $\bar{\nu}$ = 1/λ. Atomic spectra are lines; molecular spectra are bands.

FORMULAS

F1 Photon energy ★★★

$$E = h\nu$$

GIVES Energy of radiation
APPLIES any EM radiation
UNITS J

S001·5

F2 Frequency–wavelength ★★★

$$\nu = \frac{c}{\lambda}$$

GIVES Frequency
APPLIES vacuum/air
UNITS Hz

S001·5

F3 Wavenumber ★★★

$$\bar{\nu} = \frac{1}{\lambda} \qquad E = hc\bar{\nu}$$

GIVES Wavenumber; energy
APPLIES spectroscopic unit
UNITS cm^{-1}

S001·5

F4 Transition energy ★★★

$$\Delta E = E_2 - E_1 = h\nu$$

GIVES Absorbed/emitted quantum
APPLIES two quantised levels
UNITS J (÷hc → cm^{-1})

S001

KEY POINTS

DEFINITION Molecular spectroscopy: the branch of science dealing with the interaction of electromagnetic radiation with matter (molecules).

★★☆ S001·3

KEY POINT Spectrophotometer: source → monochromator (entrance slit, dispersive element, exit slit) → sample → detector → recorder.

★★☆ S001·3

KEY POINT

EM radiation has dual character (photons moving as waves); E and B fields oscillate perpendicular to each other and to the direction of propagation; all travel at the speed of light.

★★☆ S001·4

KEY POINT

Atomic spectra: electronic transitions between atomic orbitals → line spectra. Molecular spectra: electronic (HOMO→LUMO) plus vibrational and rotational transitions → band spectra.

★★☆ S001·6-7

KEY POINT

Absorption: molecule goes to a higher level taking $\Delta E = h\nu$; emission: falls to a lower level releasing $h\nu$.

★☆☆ S001

FLOWCHARTS & COMPARISONS

Atomic vs molecular spectra

	Atomic	Molecular
Transition	between atomic orbitals	HOMO→LUMO + vib + rot
Appearance	line spectra	band spectra

T2 Electromagnetic spectrum & regions ★★★

SHORT NOTE

Radio→NMR/ESR; microwave→rotation; IR→vibration; UV-Vis→valence electrons; X-ray→inner electrons; γ→nuclei. Raman: Rayleigh (no shift), Stokes (lower ν), anti-Stokes (higher ν).

KEY POINTS

KEY POINT

Radio (10 m–1 cm): NMR, ESR – nuclear / electron spin reversal. Microwave (1 cm–100 μm): rotational. IR (100 μm–1 μm): vibrational. UV-Vis (1 μm–10 nm): electronic, valence electrons. X-ray (10 nm–100 pm): inner electrons. γ (100–1 pm): nuclear rearrangement.

★★★ S002·4-5

WHY

Rayleigh scattering: scattered light has the same frequency as the incident light (elastic). Raman: shifted frequency — Stokes lines (lower frequency, molecule gains energy), anti-Stokes lines (higher frequency, molecule loses energy).

★★☆ T002 TRANSCRIPT ONLY ADDED FOR CLARITY

FLOWCHARTS & COMPARISONS

Microwave vs IR vs UV-Vis

	Microwave	IR	UV-Vis
Transition	rotational	vibrational	electronic
Condition	permanent dipole	dipole changes	any molecule
Rule	ΔJ=±1	Δv=±1 (SHO)	none for v; Franck–Condon
Gives	B, I, r_o	k, $\bar{\nu}$	electronic states

Rayleigh vs Raman

	Rayleigh	Stokes	Anti-Stokes
Frequency	unchanged	lower	higher
Molecule	unchanged	gains energy	loses energy

ASKED IN PAST ESA PAPERS

(×2) Rayleigh vs Raman; Stokes vs anti-Stokes.

T3 Born–Oppenheimer approximation ★★☆

SHORT NOTE

$E = E_{rot} + E_{vib} + E_{elec}$, $E_{rot} \ll E_{vib} \ll E_{elec}$, components independent; translation negligible.

FORMULAS

F5 Total molecular energy ★★☆

$$E = E_{trans} + E_{rot} + E_{vib} + E_{elec} \approx E_{rot} + E_{vib} + E_{elec}$$

GIVES Born–Oppenheimer
APPLIES E_{trans} negligible
UNITS J

S003·4

F6 Ordering ★★☆

$$E_{rot} \ll E_{vib} \ll E_{elec}$$

GIVES Microwave < IR < UV-Vis
UNITS —

S003·4

F7 Change in energy ★★☆

$$\Delta E_{total} = \Delta E_{elec} + \Delta E_{vib} + \Delta E_{rot}$$

GIVES Electronic transition fine structure
UNITS J or cm^{-1}

S007·4

KEY POINTS

LAW

> *Born–Oppenheimer approximation: electronic, vibrational and rotational energies of a molecule are independent;* $E = E_{rot} + E_{vib} + E_{elec}$ *(*E_{trans} *negligible).*

★★☆ S003·4

KEY POINT

$E_{rot} \ll E_{vib} \ll E_{elec}$ because electronic levels are farther apart than vibrational, which are farther than rotational.

★★☆ S003·4

EXAM TRAP

Mnemonic: Rot < Vib < Elec ↔ Microwave < IR < UV-Vis.

★☆☆ study aid ADDED FOR CLARITY

ASKED IN PAST ESA PAPERS

(×3) State Born–Oppenheimer; energy-level diagram.

T3b Beer–Lambert law ★☆☆

SHORT NOTE

A = log(I_0/I) = εcl (older syllabus; not on current slides).

FORMULAS

F8 Beer–Lambert ★☆☆

$$A = \log \frac{I_0}{I} = \varepsilon c l$$

NOT ON CURRENT SLIDES
ADDED FOR CLARITY

GIVES	Absorbance
APPLIES	dilute solutions
UNITS	dimensionless

KEY POINTS

KEY POINT

Beer–Lambert law: absorbance A = log(I_0/I) = εcl — proportional to concentration c and path length l; ε is the molar absorptivity.

★☆☆ ESA Dec-24 NOT ON CURRENT SLIDES ADDED FOR CLARITY

ASKED IN PAST ESA PAPERS

Dec-24 (older syllabus) State the Beer–Lambert law.

T4 Rotational (microwave) spectroscopy ★★★

SHORT NOTE

Rigid rotor: two masses at fixed r_0. $I = \mu r_0^2$. $\varepsilon_J = BJ(J+1)$. Needs permanent dipole; $\Delta J = \pm 1$. Lines at $2B(J+1)$, spacing $2B$. From B get I and r_0.

FORMULAS

F9 Centre of mass ★★★

$$m_1 r_1 = m_2 r_2 \qquad r_0 = r_1 + r_2$$

GIVES Positions of atoms about C
APPLIES rigid diatomic
UNITS m

S004; PDF

F10 Distances ★★★

$$r_1 = \frac{m_2\, r_0}{m_1 + m_2} \qquad r_2 = \frac{m_1\, r_0}{m_1 + m_2}$$

UNITS m

PDF

F11 Moment of inertia ★★★

$$I = m_1 r_1^2 + m_2 r_2^2 = \mu r_0^2, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

GIVES Moment of inertia
APPLIES rigid rotor
UNITS kg m²

S004·7; PDF

F12 Rotational energy ★★☆

$$E_r = \tfrac{1}{2} I\omega^2 = \frac{L^2}{2I}$$

GIVES Classical rotor
UNITS J

PDF

F13 Rotational levels (J) ★★★

$$E_J = \frac{h^2}{8\pi^2 I}\, J(J+1)\ \mathrm{J}$$

GIVES Quantised levels
APPLIES J = 0,1,2...
UNITS J

S004·8

F14 Rotational levels (cm⁻¹) ★★★

$$\varepsilon_J = \frac{h}{8\pi^2 Ic}\, J(J+1) = BJ(J+1)\ \mathrm{cm}^{-1}, \qquad B = \frac{h}{8\pi^2 Ic}$$

GIVES Level energy; rotational constant
APPLIES c in cm s⁻¹
UNITS cm⁻¹

S004·8

F15 **Selection rules** ★★★

$$\Delta J = \pm 1$$

GIVES Allowed transitions
APPLIES gross rule
UNITS —

S004·10

F16 **Line positions** ★★★

$$\Delta\varepsilon_{J\to J+1} = 2B(J+1)\ \text{cm}^{-1}; \quad \text{levels } 0, 2B, 6B, 12B, 20B, 30B, 42B$$

GIVES Rotational spectrum
UNITS cm^{-1}

S004·9-10

DERIVATION

DERIVATION · MOMENT OF INERTIA OF A DIATOMIC RIGID ROTOR · I = MR$_0^2$

1	$m_1 r_1 = m_2 r_2 = m_2(r_0 - r_1)$	Balance about the centre of mass C; $r_0 = r_1 + r_2$
2	$r_1 = \dfrac{m_2 r_0}{m_1 + m_2}, \qquad r_2 = \dfrac{m_1 r_0}{m_1 + m_2}$	Solve for each distance
3	$I = m_1 r_1^2 + m_2 r_2^2 = \dfrac{m_1 m_2^2 r_0^2 + m_2 m_1^2 r_0^2}{(m_1 + m_2)^2}$	Substitute into the definition of I
4	$I = \dfrac{m_1 m_2}{m_1 + m_2} r_0^2 = \mu r_0^2$	Factor $m_1m_2(m_1+m_2)$

KEY POINTS

DEFINITION Gross selection rule (rotational): molecule must have a permanent dipole moment. Specific rule: ΔJ = ±1.

★★★ S004·3,10

WHY HCl and CO are microwave-active (permanent dipole); H_2, Cl_2 and CO_2 are not (no permanent dipole).

★★★ S004·5

KEY POINT Rigid-rotor model: heteronuclear diatomic = two masses joined by a rigid massless rod; J = 0,1,2...; levels 0, 2B, 6B, 12B...

★★★ S004·6-9

KEY POINT First line is at 2B (not B) and consecutive lines are 2B apart; B → I → r_0 via $I = \mu r_0^2$.

★★★ S004·12

KEY POINT Intensity rises then falls with J because higher levels are populated at room temperature (thermal population).

★★☆ S004·12

EXAM TRAP Traps: J starts at 0; first line = 2B; use μ in kg (not amu); use c in cm s^{-1} when B is in cm^{-1}.

★★★ exam

GRAPHS & DIAGRAMS

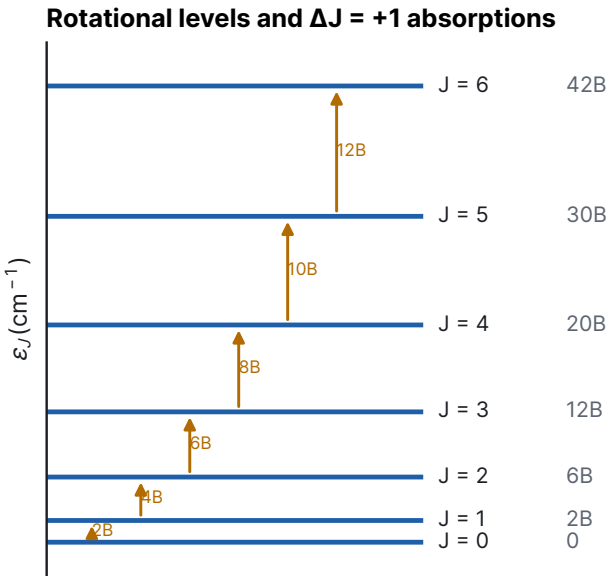


Fig. G9 Rigid-rotor energy levels J=0–6 (values J(J+1)B), ΔJ=+1 arrows.

IN THE EXAM Label J, the energy in B, and the widening gaps 2B,4B,6B...

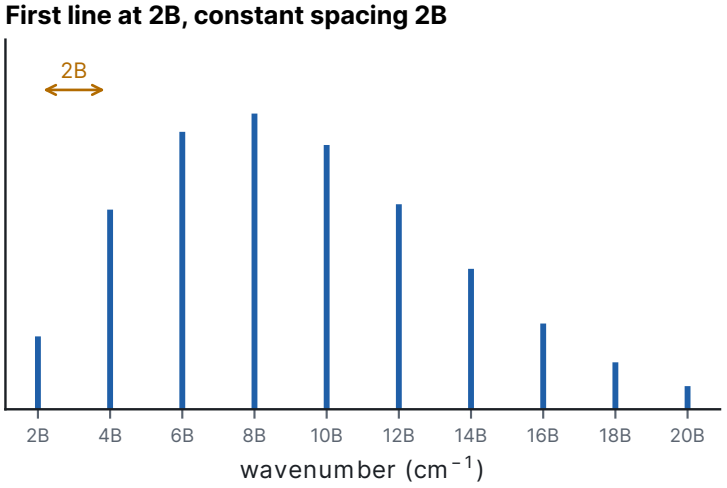


Fig. G10 Rotational stick spectrum.

IN THE EXAM Lines at 2B, 4B, 6B...; constant spacing 2B; intensity rises then falls.

FLOWCHARTS & COMPARISONS

F3 Spectroscopic activity checker

1. Permanent dipole? → Yes: microwave (rotational) active
2. Dipole changes during a vibration? → Yes: IR active
3. All molecules → electronic (UV-Vis) active
4. HCl: all three · CO: all three · H_2, Cl_2: electronic only · CO_2: IR (asym. stretch, bend) + electronic, not microwave · H_2O: all three · HI: all three

F4 Rotational spectrum → bond length

1. Read first line (or spacing) → B = line/2
2. $I = h / (8\pi^2 cB)$
3. $\mu = m_1m_2/(m_1+m_2)$ in kg
4. $r_o = \sqrt{(I/\mu)}$

WORKED EXAMPLES

WORKED EXAMPLE T4.1 — ESA

GIVEN	HCl: first rotational line at 21.18 cm^{-1} (H=1.008, Cl=35.45). Find B, I, r_0.
METHOD	B = line/2 = 10.59 cm^{-1}; $I = h/(8\pi^2cB)$ = 2.64e-47 kg m^2; μ = 1.63e-27 kg; $r_0 = \sqrt{(I/\mu)}$
ANSWER	**B = 10.59 cm^{-1}, I = 2.64e-47 kg m^2, r_o = 1.27 Å**

WORKED EXAMPLE T4.2 — ESA

GIVEN	HI: first line 13.022 cm^{-1} (H=1.0, I=126.9). Find I and r_0.
METHOD	B = 6.511 cm^{-1}; $I = h/(8\pi^2cB)$; $r_0 = \sqrt{(I/\mu)}$
ANSWER	**I = 4.30e-47 kg m^2, r_o = 1.62 Å**

WORKED EXAMPLE T4.3 — ESA

GIVEN	B = 10.56 cm^{-1}. Find I and line spacing.
METHOD	$I = h/(8\pi^2cB)$; spacing = 2B
ANSWER	**I = 2.65e-47 kg m^2; spacing = 21.12 cm^{-1}**

WORKED EXAMPLE T4.4 — ESA

GIVEN	HCl J=0→1 at 20.68 cm^{-1}. Find the J=20→21 line.
METHOD	$\Delta\varepsilon = 2B(J+1)$ and 2B = 20.68
ANSWER	**2B·21 = 21 × 20.68 = 434.28 cm^{-1}**

ASKED IN PAST ESA PAPERS

2023–25 (×4) Derive I = μr_0^2 for a diatomic rigid rotor; calculate bond length from first line (HCl 21.18 cm^{-1}; HI 13.022 cm^{-1}); draw levels to J=5/6 and the spectrum.

T5 Vibrational (IR) – SHO ★★★

SHORT NOTE

SHO: spring/Hooke's law, parabolic PE. $\varepsilon_v = (v+½)\bar{\nu}$, ZPE = ½$\bar{\nu}$, equal spacing, Δv=±1 → one line at $\bar{\nu}$. Needs changing dipole; $\bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)}$.

FORMULAS

F17 Hooke's law ★★☆

$$F = -kx \qquad PE = \tfrac{1}{2}k(r - r_{eq})^2$$

GIVES Parabolic potential
APPLIES SHO
UNITS N ; J

S005·6-7

F18 Oscillation frequency ★★★

$$\nu_{osc} = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}\ \text{Hz}$$

GIVES Frequency
APPLIES SHO
UNITS Hz

S005·8

F19 Wavenumber of vibration ★★★

$$\bar{\nu}_{osc} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}\ \text{cm}^{-1}$$

GIVES Line position
APPLIES c in cm s^{-1}
UNITS cm^{-1}

S005·8

F20 SHO levels ★★★

$$E_v = \left(v + \tfrac{1}{2}\right) h\nu_{osc}\ \text{J} \qquad \varepsilon_v = \left(v + \tfrac{1}{2}\right) \bar{\nu}_{osc}\ \text{cm}^{-1}$$

GIVES Levels
APPLIES v = 0,1,2...
UNITS J ; cm^{-1}

S005·8-9

F21 **Zero-point energy** ★★★

$$\varepsilon_0 = \tfrac{1}{2}\,\bar{\nu}_{osc}$$

GIVES Lowest level (≠ 0)
UNITS cm^{-1}

S005·9

F22 **Selection rule & spacing** ★★★

$$\Delta v = \pm 1; \quad \varepsilon_{v+1} - \varepsilon_v = \bar{\nu}_{osc} \;\Rightarrow\; \text{one line at } \bar{\nu}_{osc}$$

APPLIES dipole must change
UNITS cm^{-1}

S005·9

F23 **Force constant** ★★★

$$k = 4\pi^2 c^2 \bar{\nu}^2 \mu$$

GIVES Bond stiffness
APPLIES μ in kg
UNITS N m^{-1}

derived

KEY POINTS

DEFINITION Gross selection rule (IR): the vibration must change the dipole moment; Δv = ±1.
★★★ S005·3,9

WHY CO_2 has no permanent dipole (microwave-inactive) but is IR-active: asymmetric stretch and bending change the dipole; symmetric stretch does not.
★★★ S005·4-5

WHY HI is IR-active but H_2 and I_2 are not (homonuclear: no dipole change).
★★☆ S005·4

WHY SHO shows only one line: levels are equally spaced and Δv = ±1, so every allowed transition has the same energy $\bar{\nu}_{osc}$.
★★★ S005·9

KEY POINT Zero-point energy $\varepsilon_0 = ½\bar{\nu}_{osc} \neq 0$: the molecule never rests even at v = 0. Bond treated as a spring obeying Hooke's law → parabolic PE.
★★★ S005·6-9

KEY POINT Every bond has a characteristic frequency, so an IR spectrum is a molecular 'fingerprint'.
★☆☆ S005·6

GRAPHS & DIAGRAMS

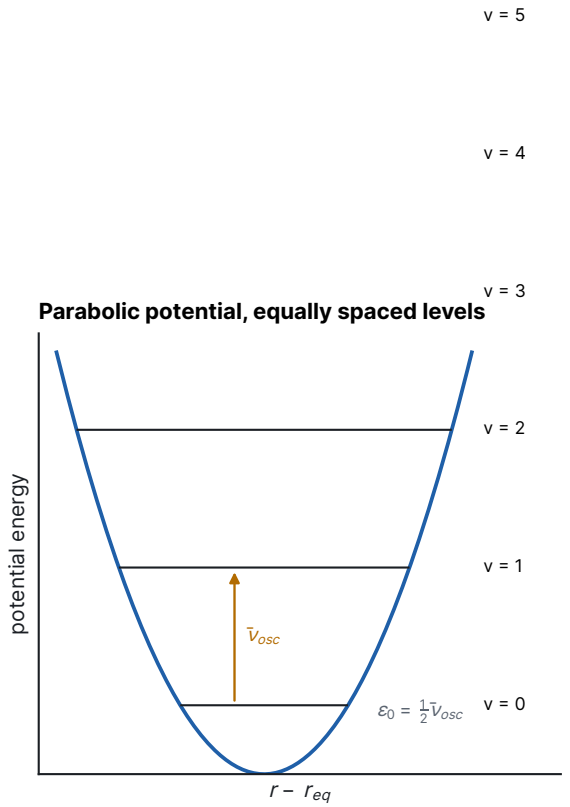


Fig. G13 SHO parabola with equally spaced levels.

IN THE EXAM Mark v=0…5 and $\varepsilon_0 = \frac{1}{2}\bar{\nu}$; spectrum is a single line.

FLOWCHARTS & COMPARISONS

F5 IR line → force constant (and reverse)

1. $\bar{\nu} \rightarrow k = 4\pi^2 c^2 \bar{\nu}^2 \mu$
2. $k \rightarrow \bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)} \rightarrow ZPE = \frac{1}{2}\bar{\nu} \rightarrow \varepsilon_v = (v+\frac{1}{2})\bar{\nu}$

WORKED EXAMPLES

WORKED EXAMPLE T5.1 ESA

GIVEN	HF, k = 970 N m^{-1} (H=1.008, F=18.99). Find $\bar{\nu}$ and zero-point energy.
METHOD	$\bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)}$; ZPE = $\frac{1}{2}\bar{\nu}$ (× hc for joules)
ANSWER	**$\bar{\nu}$ = 4148 cm^{-1}; ZPE = 2074 cm^{-1} = 4.12e-20 J**

WORKED EXAMPLE T5.2 ESA

GIVEN	CO, $\bar{\nu}$ = 2168 cm^{-1} (SHO). Find k and ZPE.
METHOD	$k = 4\pi^2c^2\bar{\nu}^2\mu$
ANSWER	**k = 1.90e+03 N m^{-1}; ZPE = 1084 cm^{-1}**

WORKED EXAMPLE T5.3 ESA

GIVEN	NO, k = 1530 N m^{-1} (N=14, O=16). Find $\bar{\nu}$, ZPE, ε_3, ε_4.
METHOD	$\varepsilon_v = (v+½)\bar{\nu}$
ANSWER	**$\bar{\nu}$ = 1865; ZPE = 933; ε_3 = 6528; ε_4 = 8393 cm^{-1}**

ASKED IN PAST ESA PAPERS

2023–25 Calculate $\bar{\nu}$ and ZPE from k (HF, NO); k from $\bar{\nu}$ (CO); why only one line in SHO?

T6 Vibrational – anharmonic ★★☆

SHORT NOTE

Morse potential: $\varepsilon_v = (v+½)\bar{\omega_e} - (v+½)^2\bar{\omega_e}\bar{x_e}$. Levels converge; Δv=±1,±2...; fundamental, overtones; hot bands left of fundamental.

FORMULAS

F24 Morse function ★★☆

$$PE = D_{eq}\left[1 - \exp\{a(r_{eq} - r)\}\right]^2$$

GIVES Anharmonic potential
UNITS J

S006·3

F25 Anharmonic levels ★★☆

$$\varepsilon_v = \left(v + \tfrac{1}{2}\right)\bar{\omega}_e - \left(v + \tfrac{1}{2}\right)^2 \bar{\omega}_e x_e \ \mathrm{cm}^{-1}$$

GIVES Converging levels
APPLIES x_e small, positive
UNITS cm^{-1}

S006·4

F26 Transitions ★★☆

$$0 \to 1: \ \bar{\omega}_e(1 - 2x_e) \quad \text{fundamental}$$
$$0 \to 2: \ 2\bar{\omega}_e(1 - 3x_e) \quad \text{1st overtone}$$
$$0 \to 3: \ 3\bar{\omega}_e(1 - 4x_e) \quad \text{2nd overtone}$$

APPLIES Δv = ±1,±2,±3...
UNITS cm^{-1}

S006·6-7

F27 **Hot band & ZPE** ★★☆

$$1 \rightarrow 2: \ \bar{\omega}_e(1-4x_e) \qquad \varepsilon_0 = \tfrac{1}{2}\bar{\omega}_e\left(1-\tfrac{1}{2}x_e\right)$$

ADDED FOR CLARITY

APPLIES higher T
UNITS cm^{-1}

S006·9; ZPE added

KEY POINTS

WHY

Why SHO fails: real bonds dissociate when stretched; the true potential is anharmonic (Morse), so levels converge and overtones appear.

★★☆ S006·3

KEY POINT

Anharmonic: Δv = ±1, ±2, ±3...; most transitions start from v = 0 → fundamental (0→1), first overtone (0→2), second overtone (0→3).

★★☆ S006·6

KEY POINT

Hot bands: from v = 1 at higher temperature; 1→2 appears to the left of the fundamental.

★★☆ S006·9

KEY POINT

x_e is small and positive; line spacing decreases with increasing v.

★★☆ S006·4

GRAPHS & DIAGRAMS

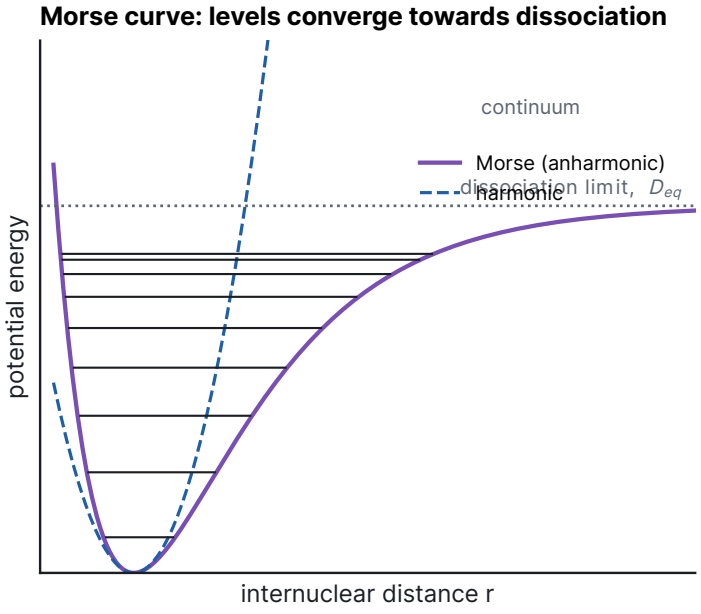


Fig. G15 Morse curve vs harmonic parabola.

IN THE EXAM Show D_{eq}, convergence of levels and dissociation limit.

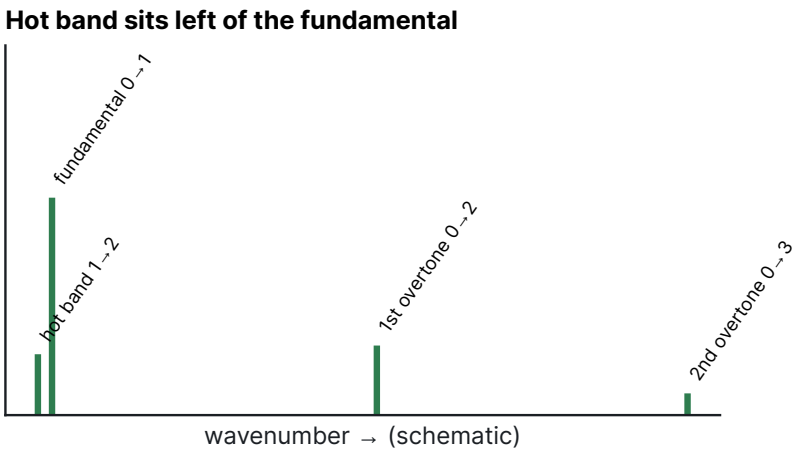


Fig. G17 Anharmonic IR spectrum (schematic).

IN THE EXAM Hot band left of fundamental; overtones with decreasing spacing.

FLOWCHARTS & COMPARISONS

F6 **Anharmonic solver**

1. fundamental = $\bar{\omega}_e(1-2x_e)$; overtone = $2\bar{\omega}_e(1-3x_e)$
2. ratio overtone/(2·fund) = $(1-3x_e)/(1-2x_e)$ → x_e
3. $\bar{\omega}_e$ = fundamental/$(1-2x_e)$

SHO vs anharmonic

	SHO	Anharmonic
Potential	parabola	Morse
Levels	equal spacing	converge
Rule	Δv=±1	Δv=±1,±2,±3…
Spectrum	1 line	fundamental + overtones + hot bands

WORKED EXAMPLES

WORKED EXAMPLE T6.1 ESA

GIVEN	HBr: fundamental 2559.08, first overtone 5027.54 cm^{-1}. Find x_e and $\bar{\omega_e}$.
METHOD	fund = $\bar{\omega_e}(1-2x_e)$; overtone = $2\bar{\omega_e}(1-3x_e)$
ANSWER	**x_e = 0.0171; $\bar{\omega}_e$ = 2649.7 cm^{-1}**

WORKED EXAMPLE T6.2 ESA

GIVEN	CO: fundamental 2143 cm^{-1}, $x_e = 0.01$. Find $\bar{\omega}_e$.
METHOD	$\bar{\omega}_e$ = fund/(1−2x_e)
ANSWER	**$\bar{\omega}_e$ = 2186.7 cm^{-1}**

WORKED EXAMPLE T6.3 ESA

GIVEN	CO: $\bar{\omega}_e = 2050$, $x_e = 0.017$. First overtone?
METHOD	$2\bar{\omega}_e(1-3x_e)$
ANSWER	**3890.9 cm^{-1}**

ASKED IN PAST ESA PAPERS

2023–25 Find x_e and $\bar{\omega}_e$ from fundamental and overtone; selection rule; failure of SHO.

T7 Electronic spectroscopy & Franck–Condon ★★★

SHORT NOTE

HOMO → LUMO transitions; vibrational coarse structure gives progressions (v′,v″); no vibrational selection rule; Franck–Condon: vertical transitions → intensity pattern by r_e' vs r_e''; continuum = dissociation.

FORMULAS

F28 Electronic spectrum ★★☆

$$\varepsilon = \varepsilon_{elec} + \varepsilon_{vib}; \quad \text{lines labelled } (v', v'')$$

GIVES Progression
APPLIES no vibrational selection rule
UNITS cm^{-1}

S007·6-8

KEY POINTS

KEY POINT Electronic transitions: HOMO → LUMO (σ → σ*, π → π*, n → π*) in the UV-Vis; all molecules show them; visible transitions give colour.
★★★ S007·3

KEY POINT Vibrational changes give 'coarse structure', rotational changes give 'fine structure'. Homonuclear molecules (H_2, N_2) show vibration/rotation only through electronic spectra.
★★☆ S007·5

KEY POINT

No vibrational selection rule during electronic transitions; most start at v″ = 0; lines labelled (v′,v″) form a progression converging at high energy (anharmonic excited state).

★★★ S007·6-8

LAW

> *Franck–Condon principle: "An electronic transition takes place so rapidly that a vibrating molecule does not change its internuclear distance appreciably during the transition." Hence transitions are vertical.*

★★★ S007·13

KEY POINT

Case (a) $r_e' = r_e''$: (0,0) strongest. Case (b) $r_e' > r_e''$: strongest line (v′,0) with v′>0. Case (c) $r_e' \gg r_e''$: transition goes beyond the dissociation limit → continuum.

★★★ S007·15-16

WHY

A continuum means dissociation: the atoms fly apart with unquantised kinetic energy. The excited state usually has a longer bond length because of antibonding character.

★★★ S007·10,16

GRAPHS & DIAGRAMS

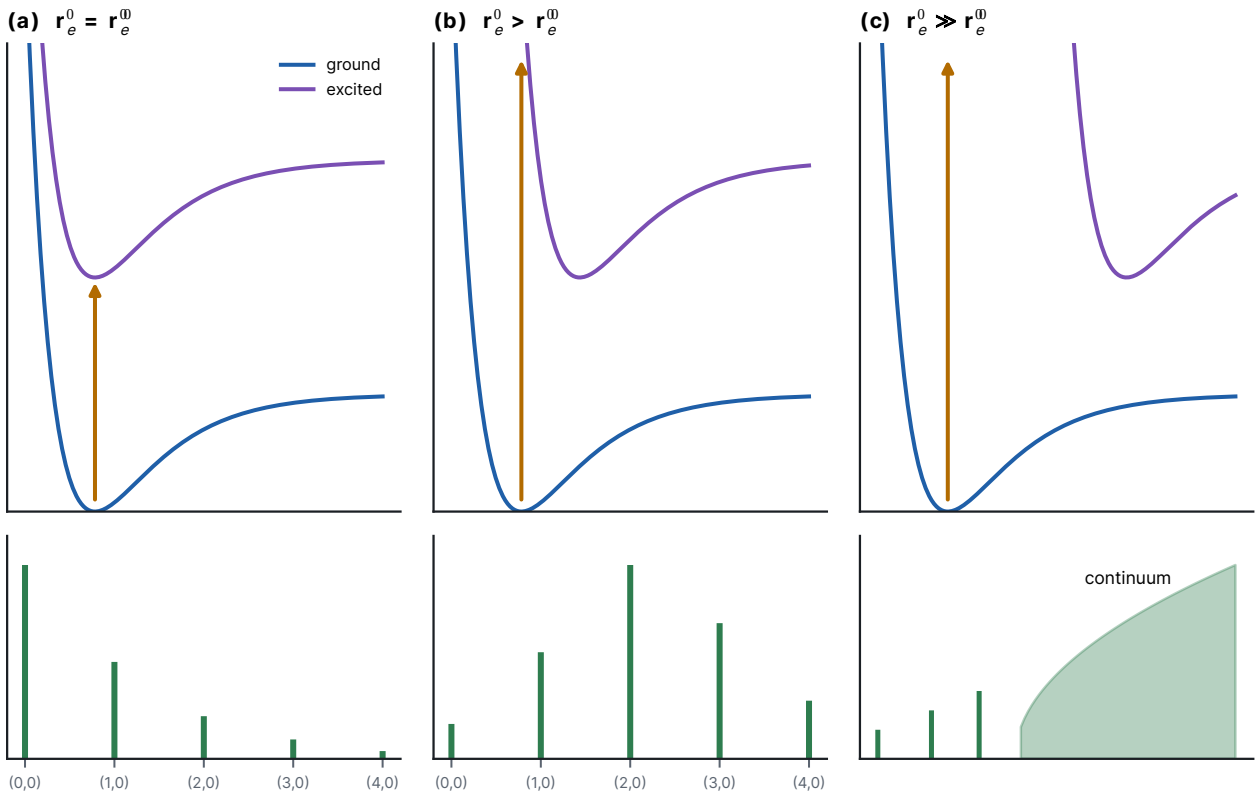


Fig. G21 Franck–Condon, three cases (a, b, c) with spectra.

IN THE EXAM Draw vertical arrow from v″=0; (c) ends in a continuum.

FLOWCHARTS & COMPARISONS

F7 Franck–Condon outcome

1. Compare r_e' with r_e''
2. Equal → (0,0) strongest
3. Longer → (v′,0), v′>0 strongest
4. Much longer → continuum (dissociation)

ASKED IN PAST ESA PAPERS

2023–25 (×5) State the Franck–Condon principle; draw the three cases with spectra; meaning of a continuum.

MODULE 2

Phase Equilibria

T8 · Basic terms & phase rule | T9 · Derivation of phase rule | T10 · One-component system: water | T11 · Two-component: reduced phase rule & Pb–Ag | T12 · Thermal analysis / cooling curves

T8 Basic terms & phase rule ★★★

SHORT NOTE

Phase = homogeneous, separable part; component = min. independent constituents; F = min. intensive variables. F = C–P+2.

FORMULAS

F29 Phase rule ★★★

$$F = C - P + 2$$

GIVES Degrees of freedom
APPLIES no external fields
UNITS —

S008·12

KEY POINTS

DEFINITION Phase (P): homogeneous, physically distinct, mechanically separable part of a system bounded by a surface. Gases: P = 1; miscible liquids: P = 1; each solid / polymorph is a phase.
★★★ S008·4

DEFINITION Component (C): smallest number of independently variable constituents by which the composition of every phase can be expressed (zero/negative values allowed).
★★★ S008·6

DEFINITION Degrees of freedom (F): minimum number of intensive variables (T, P, concentration) to be fixed to define the system completely.
★★★ S008·10

LAW

> *Gibbs phase rule: provided equilibrium is not influenced by external forces (gravity, electrical, magnetic), F = C − P + 2.*

★★★ S008·12

KEY POINT Counting: ice/water/vapour P=3, C=1; $CaCO_3 \rightleftharpoons CaO + CO_2$: P=3, C=2; $NH_4Cl \rightleftharpoons NH_3 + HCl$: P=2, C=1 (C=2 if extra HCl/NH_3 added); $Na_2SO_4 \cdot 10H_2O$ system C=2.
★★☆ S008·5-9

KEY POINT Systems minimise Gibbs energy at constant T, P; at equilibrium G is equal in coexisting phases.

★☆☆ S008·3

FLOWCHARTS & COMPARISONS

F9 **Phase-rule application**

1. Count P
2. Count C (minimum constituents)
3. Gas absent & constant pressure? → F=C−P+1, else F=C−P+2
4. Classify F=2 bivariant, 1 univariant, 0 invariant

ASKED IN PAST ESA PAPERS

(×2) Components and phases for $CaCO_3$/$MgCO_3$ decomposition.

T9 Derivation of phase rule ★★★

SHORT NOTE

Variables P(C−1)+2, constraints C(P−1), equal μ in all phases → F = C−P+2.

FORMULAS

F30 **Variables & constraints** ★★★

$$F = \underbrace{P(C-1)+2}_{\text{variables}} - \underbrace{C(P-1)}_{\text{constraints}} = C - P + 2$$

GIVES Derivation
UNITS —

S009·7

F31 **Equilibrium condition** ★★★

$$\mu_i^{\alpha} = \mu_i^{\beta} = \mu_i^{\gamma} = \cdots$$

GIVES Equal chemical potential
APPLIES each component
UNITS J mol^{-1}

S009·3,7

F32 **Chemical potential** ★★☆

$$\mu_i = \left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_j}$$

TRANSCRIPT ONLY

GIVES Partial molar Gibbs energy
UNITS J mol^{-1}

T008 (S009·5 unreadable)

DERIVATION

DERIVATION · GIBBS PHASE RULE FOR C COMPONENTS AND P PHASES

1	composition variables $= P(C-1)$	Each phase needs C − 1 mole fractions (they sum to 1)
2	total variables $= P(C-1)+2$	Add temperature and pressure (same in every phase)
3	$\mu_i^{\alpha} = \mu_i^{\beta} = \cdots \Rightarrow (P-1)$ equations per component	Equilibrium: equal chemical potential in all phases
4	total constraints $= C(P-1)$	Over all C components
5	$F = P(C-1)+2-C(P-1) = C-P+2$	Degrees of freedom = variables − constraints

KEY POINTS

KEY POINT At equilibrium all phases have the same T and P and the chemical potential of each component is equal in every phase; mass flows from higher to lower μ.

★★★ S009·3-4

KEY POINT Derivation: variables = P(C−1) composition terms + 2 (T,P); constraints = C(P−1); F = C−P+2.

★★★ S009·7

FLOWCHARTS & COMPARISONS

F8 **Phase-rule derivation**

1. Variables = P(C−1)+2
2. Constraints = C(P−1)
3. F = variables − constraints = C−P+2

ASKED IN PAST ESA PAPERS

2023–25 (×3) Derive the phase rule for C components and P phases.

T10 One-component system: water ★★★

SHORT NOTE

F = 3–P. Areas ice/water/vapour F=2; lines OA, OB, OC F=1; triple point O F=0 (0.0098 °C, 4.58 mmHg); critical 374 °C, 218 atm; OC slope negative; OA′ metastable.

FORMULAS

F33 One-component ★★★

$$F = 3 - P$$

GIVES Area F=2, line F=1, point F=0
UNITS —

S009·8

F34 Clausius–Clapeyron ★★★

$$\frac{dP}{dT} = \frac{\Delta H}{T\,\Delta V}$$

GIVES Slope of curve
APPLIES fusion, vaporisation, sublimation
UNITS Pa K^{-1}

S009·12-14

KEY POINTS

KEY POINT Water: F = 3–P → one phase F=2 (bivariant area), two phases F=1 (univariant line), three phases F=0 (invariant point).

★★★ S009·8-10

KEY POINT Triple point O: 0.0098 °C, 4.58 mmHg (F=0). Critical point A: 374 °C, 218 atm; above it water cannot be liquid.

★★★ S009·15

WHY Fusion curve OC has negative slope: ice melts with decrease in volume (ΔV<0 → dP/dT<0), and it is steep because |ΔV| is small. Sublimation (OB) and vaporisation (OA) slopes are positive and less steep because ΔV is large.

★★★ S009·12-14; T009b TRANSCRIPT-ASSISTED

KEY POINT Metastable OA′: supercooled water; its vapour pressure is higher than that of ice at the same T; disturbance returns the system to stable OB.

★★★ S009·16

GRAPHS & DIAGRAMS

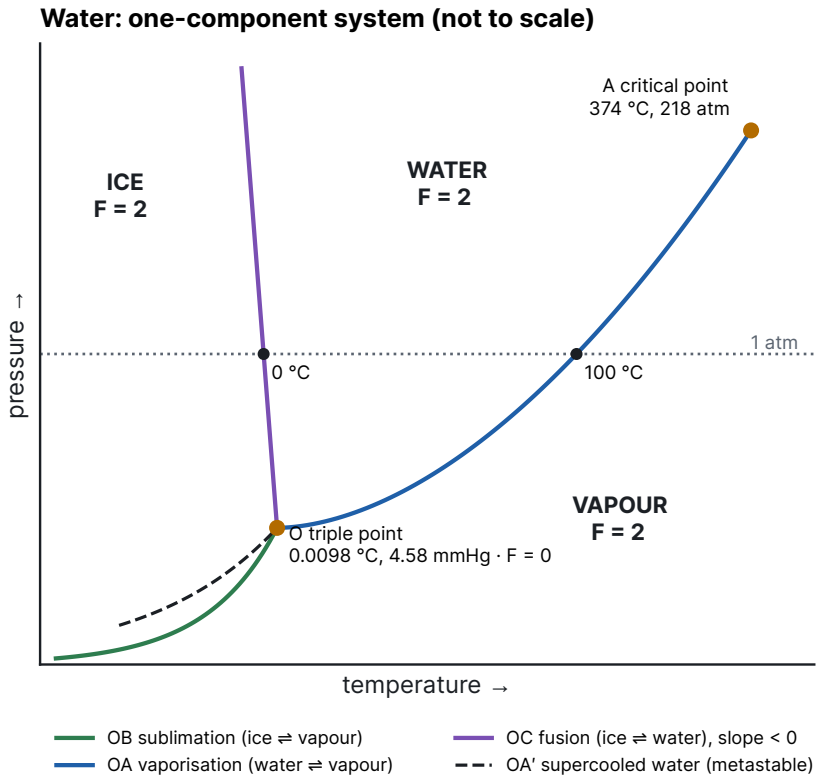


Fig. G22 Water phase diagram (schematic, not to scale).

IN THE EXAM Label OA, OB, OC, OA′, O, A, with T,P values and F in each region.

ASKED IN PAST ESA PAPERS

2023–25 (×4) Draw and label the water phase diagram; F in each region; metastable curve; negative slope of melting curve.

T11 Two-component: reduced phase rule & Pb–Ag ★★★

SHORT NOTE

Condensed system at fixed P: F = C−P+1. Pb–Ag eutectic: 303 °C, 2.6 % Ag, F=0; curves F=1.

FORMULAS

F35 **Reduced (condensed) rule** ★★★

$$F = C - P + 1$$

GIVES Constant pressure
APPLIES condensed systems
UNITS —

S010·2-3

KEY POINTS

WHY

General phase rule not used for 2 components: C=2 gives F=4−P, F=3 for P=1, needing 3-D; so pressure is fixed (≈1 atm) → F = C−P+1.

★★★ S010·2-3

DEFINITION

Condensed system: gas phase absent, hardly affected by small pressure changes.

★★☆ S010·3

KEY POINT

Pb–Ag: components miscible in all proportions in melt, immiscible as solids; plot is temperature vs composition at constant pressure.

★★★ S010·4

KEY POINT

Freezing-point curves: C=2, P=2 → F=1 (univariant). Eutectic point O: P=3, F=0 (invariant), lowest freezing point, definite composition and sharp melting point.

★★★ S010·7-8

KEY POINT

Numerical facts: Pb mp 327 °C, Ag mp 961 °C, eutectic 303 °C at 97.4 % Pb / 2.6 % Ag.

★★★ S010; PDF

KEY POINT

Desilverisation of argentiferous lead (Pattinson-type process) exploits this: Pb crystallises first, leaving Ag-rich melt.

★☆☆ T010 TRANSCRIPT ONLY

GRAPHS & DIAGRAMS

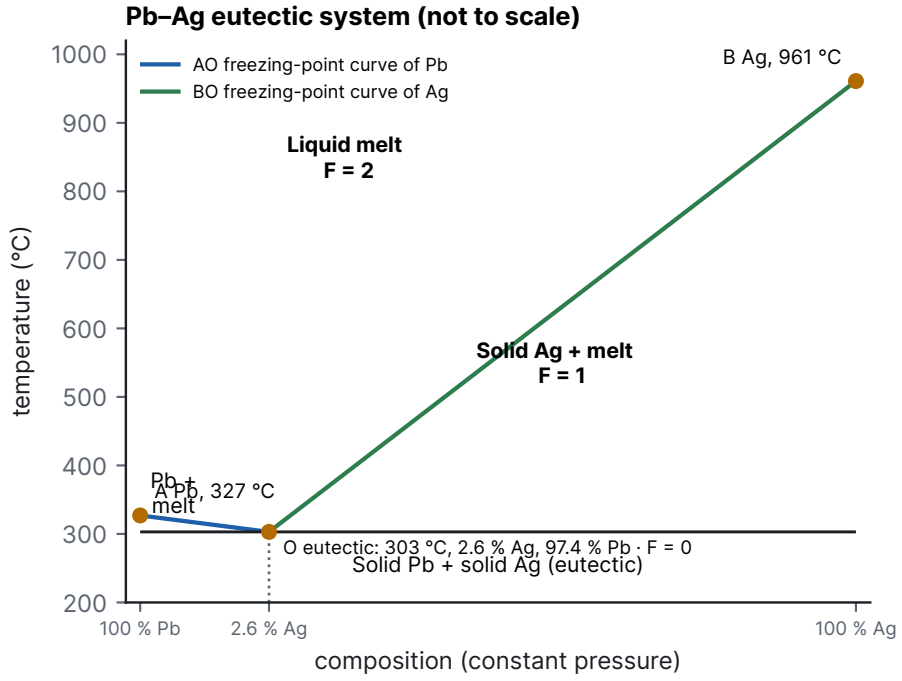


Fig. G23 Pb–Ag eutectic diagram (schematic; Pb left, Ag right).

IN THE EXAM Eutectic O at 303 °C, 2.6 % Ag; label the four areas. Slide lettering is inconsistent (see conflicts) – follow your faculty's.

FLOWCHARTS & COMPARISONS

F10 **Pb–Ag cooling path**

1. Melt cools → Pb separates, melt gets richer in Ag along curve to O
2. At O (303 °C): Pb + Ag + melt, F=0
3. Below O: solid Pb + solid Ag

ASKED IN PAST ESA PAPERS

2023–25 Draw the Pb–Ag diagram with eutectic values; why F = C−P+1.

T12 Thermal analysis / cooling curves ★★☆

SHORT NOTE

Cooling curves: halt for pure/eutectic, slope break for mixtures; plot to build phase diagram.

KEY POINTS

DEFINITION Thermal analysis: study of cooling curves (temperature vs time) of various compositions during solidification; freezing and eutectic points are read from them.

★★☆ S012·2

KEY POINT Pure substance: smooth fall, then a flat halt at the freezing point. Mixture: a change of slope when crystallisation starts, then a halt at the eutectic temperature. Plotting break/halt temperatures against composition builds the phase diagram.

★★☆ S012·2-3

GRAPHS & DIAGRAMS

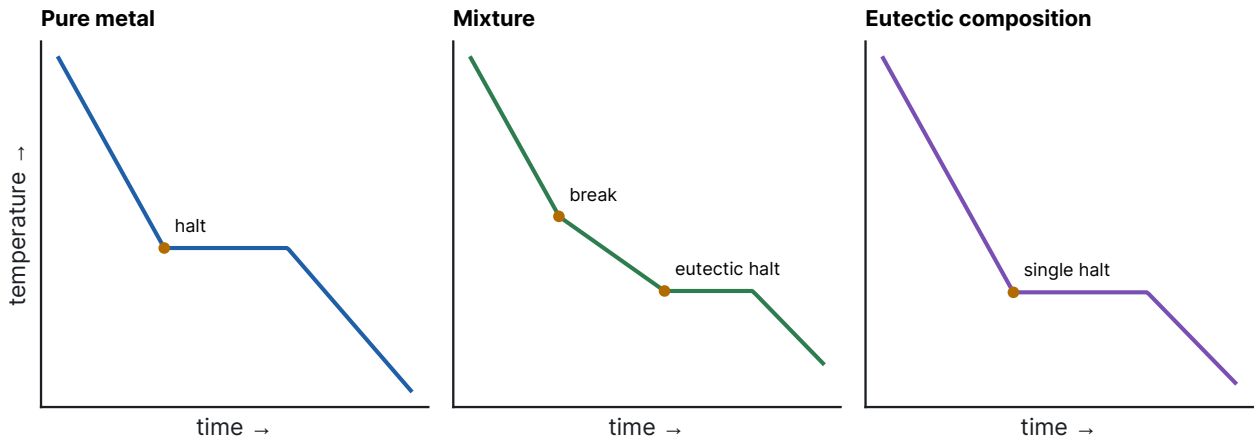


Fig. G24 Cooling curves: pure, mixture, eutectic.

IN THE EXAM Mark break and halt.

FLOWCHARTS & COMPARISONS

F11 **Thermal analysis → phase diagram**

1. Cool melts of several compositions
2. Record T vs time
3. Read break & halt temperatures
4. Plot vs composition → diagram

MODULE 3

Computational Chemistry

T13 · Computational chemistry

T13 Computational chemistry ★☆☆

SHORT NOTE

Computational chemistry simulates molecules: MM/MD (classical), semi-empirical, ab initio, DFT (quantum). Cheap, safe; AI adds speed.

KEY POINTS

DEFINITION Computational chemistry: techniques to solve chemical problems by simulating chemical systems, giving atomic-level information, using theoretical chemistry in efficient programs.

★☆☆ S011·3

KEY POINT First theoretical work: Heitler and London, 1927. (Nobel 1998 Kohn & Pople; 2013 Karplus, Levitt & Warshel.)

★☆☆ S011·3; brief ADDED FOR CLARITY

KEY POINT Scope: molecular properties (HOMO/LUMO, UV/IR/NMR spectra), reaction mechanism and kinetics, material design, drug discovery.

★☆☆ S011·6

KEY POINT Classical: molecular mechanics (balls and springs, very fast) and molecular dynamics (motion of atoms). Quantum: semi-empirical, ab initio (wavefunction) and DFT (electron density; faster than ab initio, slower than semi-empirical).

★☆☆ S011·7-12

KEY POINT Advantages: easier, cheaper, safer than experiments; complements, not replaces, experiment.

★☆☆ S011·13

KEY POINT Ethylene: sp^2 C; σ + π bond; HOMO = π, LUMO = π*. Steps: geometry → basis set → method → solve → visualise.

★☆☆ S011·14-17

KEY POINT AI-enhanced: fast screening, property prediction, reaction-pathway mapping, structure generation; speed-up from hours/days to milliseconds after training.

★☆☆ S011·19-20

GRAPHS & DIAGRAMS

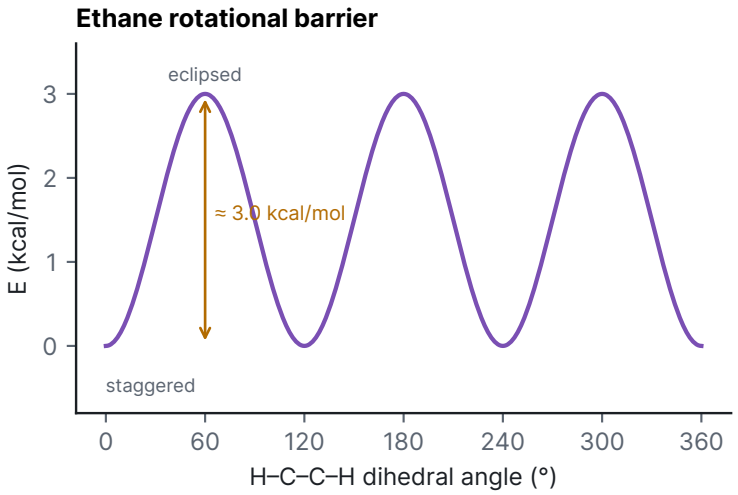


Fig. G25 Ethane torsion barrier ≈3.0 kcal/mol (illustrative).

IN THE EXAM Staggered minima, eclipsed maxima.

FLOWCHARTS & COMPARISONS

F12 Computational methods tree

1. Classical: molecular mechanics, molecular dynamics (fast, lower accuracy)
2. Quantum: semi-empirical → DFT → ab initio (slower, more accurate)

F13 Ethylene electronic structure

1. Define geometry
2. Select basis set
3. Apply method
4. Solve equations → wavefunction/density
5. Visualise HOMO, LUMO

F14 AI-enhanced workflow

1. Quantum simulations
2. Dataset
3. Train ML model
4. Predict properties in ms

MM / MD / semi-empirical / ab initio / DFT

Method	Basis	Speed	Accuracy
MM	classical springs	fastest	lowest
MD	laws of motion	fast	low
Semi-empirical	Schrödinger, approximated	fast	medium
DFT	electron density	medium	high
Ab initio	wavefunction	slowest	highest

ASKED IN PAST ESA PAPERS

(×1) Define computational chemistry; three applications.

Appendix A · Symbol glossary

Same letter, different meaning. μ is reduced mass (spectroscopy) and chemical potential (phase rule). P is number of phases and pressure. ε is energy in cm^{-1} and molar absorptivity (Beer–Lambert).

Symbol	Meaning in this unit	Unit	Value / note
λ	wavelength	m, nm, Å	
ν	frequency	Hz	
$\bar{\nu}$	wavenumber	cm^{-1}	
c	speed of light	$m\ s^{-1}$ / $cm\ s^{-1}$	3×10^{8} $m\ s^{-1}$ = 3×10^{10} $cm\ s^{-1}$
h	Planck constant	J s	6.626×10^{-34} (slides also use 6.625)
E, ε	energy; ε = energy in cm^{-1} (also molar absorptivity in Beer–Lambert — 3rd meaning!)	J; cm^{-1}; L mol^{-1} cm^{-1}	
ΔE	energy change	J	
J	rotational quantum number	—	0,1,2…
B	rotational constant = $h/8\pi^2Ic$	cm^{-1}	
I	moment of inertia	$kg\ m^2$	
μ	reduced mass $m_1m_2/(m_1+m_2)$ — ALSO chemical potential in T8–T9	kg; J mol^{-1}	1 amu = 1.66×10^{-27} kg
m_1, m_2	atomic masses	kg	
r_0	bond length (rigid rotor)	m, Å	1 Å = 10^{-10} m
r_1, r_2	distances of atoms from centre of mass	m	
ω	angular velocity	$rad\ s^{-1}$	
L	angular momentum	$kg\ m^2\ s^{-1}$	
k	force constant	$N\ m^{-1}$	
x	displacement	m	
v	vibrational quantum number	—	0,1,2…
ν_{osc}, $\bar{\nu}_{osc}$	oscillation frequency / wavenumber	Hz ; cm^{-1}	
$\bar{\omega_e}$	equilibrium vibrational wavenumber (anharmonic)	cm^{-1}	
x_e	anharmonicity constant	—	small, positive
D_{eq}	depth of Morse well	J	

Symbol	Meaning in this unit	Unit	Value / note
a	Morse constant	m^{-1}	
r_{eq}	equilibrium bond length	m	
r_e', r_e''	bond length in excited / ground state	Å	
v', v''	vibrational number in excited / ground state	—	
$\varepsilon', \varepsilon''$	energy in excited / ground state	cm^{-1}	
F	degrees of freedom (variance)	—	
C	number of components	—	
P	number of phases — ALSO pressure (clash!)	— ; atm, mmHg	
T	temperature	K, °C	
G	Gibbs energy	J	
n_i	moles of component i	mol	
ΔH	enthalpy of transition	J mol^{-1}	
ΔV	volume change of transition	m^3 mol^{-1}	
A	absorbance	—	
I_0, I	incident / transmitted intensity	—	
c, l	concentration, path length (Beer–Lambert)	mol L^{-1}, cm	
N_A	Avogadro constant	mol^{-1}	6.022×10^{23}

Appendix B · Constants & unit conversions

h	6.626×10^{-34} J s	Planck constant · slides also use 6.625
c	3×10^{8} m s^{-1} = 3×10^{10} cm s^{-1}	speed of light · use cm s^{-1} with cm^{-1}
N_A	6.022×10^{23} mol^{-1}	Avogadro constant
amu	1.66×10^{-27} kg	atomic mass unit · μ must be in kg
π	3.14	pi · as used in papers

- J → cm^{-1} **divide by hc**
- cm^{-1} → Hz **multiply by c**
- 1 eV **1.602×10^{-19} J = 8065.5 cm^{-1}**
- 1 kJ mol^{-1} **83.59 cm^{-1}**
- 1 Å **10^{-10} m**

Appendix C · Possibly examinable, not in current material

NOT ON CURRENT SLIDES

Iron–carbon (Fe–C) phase diagram. Named in the 2025 module list and asked in 2024 ESAs (peritectic and eutectoid points). The class 010 transcript says it comes “next class”, but there is no 2026 deck or transcript for it.

NOT ON CURRENT SLIDES

CO_2 phase diagram. Asked once (January 2024); not in the current material.

NOT ON CURRENT SLIDES

Beer–Lambert law. In the 2025 module overview and the Dec 2024 ESA; summarised under T3b.

Appendix D · Sources & known gaps

Source	Used for
12 class slide decks (2026), rendered .pptx + .slides.md/.txt	Definitions, statements, formulas, values
Unit-1 slide PDF (2025, 137 pp.)	Cross-check of derivations
ESA past papers (Q1), via the brief's ranking	Exam weights, worked examples

Known gaps. The class 003 video failed to download (HTTP 401), so it has no transcript. Class 012 never had a video. No videos were consulted.